Spectral-Genus Conjecture Calculation SOP

Goal for the Week: Manually calculate the spectral eigenvalues of D_4 to verify the conjecture.

I. Core Logical Chain

$$\text{Select } (G,S) \xrightarrow{\text{Decomposition}} A = \bigoplus_{\rho \in \hat{G}} \deg(\rho) \cdot A_\rho \xrightarrow{\text{Calculate Eigenvalues}} \text{Spec}(A) \xrightarrow{\text{Comparison}} \text{Surface Type}$$

Key Idea: There is no need to brute-force the $|G| \times |G|$ large matrix. Simply calculate the small matrix A_ρ for each irreducible representation ρ, then aggregate the eigenvalues.

II. Key Definitions

Definition of A_ρ

$$A_\rho = \sum_{s \in S} \rho(s)$$

A_ρ is the "adjacency operator as seen in representation ρ." For each generator $s \in S$, simply sum up the $\rho(s)$ matrices.

Eigenvalue Multiplicity Formula

$$\text{Multiplicity in } A = \deg(\rho) \times \text{Multiplicity of said eigenvalue in } A_\rho$$

Because each ρ appears $\deg(\rho)$ times in the regular representation, the eigenvalues of A_ρ are "amplified."

Composition of the Full Spectrum

$$\text{Spec}(A) = \bigcup_{\rho \in \hat{G}} \{\text{Spec}(A_\rho),\ \text{each multiplicity } \times \deg(\rho)\}$$

Laplacian Eigenvalue Conversion

$$\mu_i = |S| - \lambda_i$$

Since $L = |S| \cdot I - A$, and $|S| = 3$ for this example.

III. Basic Data for D_4

- **Group:** $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}, \quad |D_4| = 8$
- **Generating Set:** $S = \{r, r^{-1}, s\} = \{r, r^3, s\}, \quad |S| = 3$
- **Result:** $V = 8, E = 12$, the Cayley graph is 3-regular.

IV. Character Table for D_4

D_4 has 5 conjugacy classes, thus 5 irreducible representations:

	e	r^2	$\{r, r^3\}$	$\{s, r^2s\}$	$\{rs, r^3s\}$
ρ_1	1	1	1	1	1

ρ_2	1	1	1	-1	-1
ρ_3	1	1	-1	1	-1
ρ_4	1	1	-1	-1	1
ρ_5	2	-2	0	0	0

Dimension Check: $1^2 + 1^2 + 1^2 + 1^2 + 2^2 = 8 = |D_4|$ ✓

V. Calculating A_{ρ_i} (1D Representations)

For 1D representations, $\rho_i(g) \in \mathbb{C}$ is a scalar value:

$$A_{\rho_i} = \rho_i(r) + \rho_i(r^{-1}) + \rho_i(s) \in \mathbb{C}$$

Read $\rho_i(r)$ and $\rho_i(s)$ from the character table. Note that $\rho_i(r^{-1}) = \rho_i(r)$ as r^{-1} and r are in the same conjugacy class:

Rep	$\rho_i(r)$	$\rho_i(r^{-1})$	$\rho_i(s)$	A_{ρ_i} Eigenvalue	Multiplicity in A
ρ_1	1	1	1	**3**	1
ρ_2	1	1	-1	**1**	1
ρ_3	-1	-1	1	**-1**	1
ρ_4	-1	-1	-1	**-3**	1

VI. Calculating A_{ρ_5} (2D Representation)

6-1. Matrix Setup

The standard 2D representation of D_4 (rotation + reflection):

$$\rho_5(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \rho_5(r^{-1}) = \rho_5(r)^T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \rho_5(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

> **Why $\rho_5(r^{-1}) = \rho_5(r)^T$?** Rotation matrices are orthogonal, so the inverse equals the transpose.

6-2. Summation

$$A_{\rho_5} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Notice the first two terms cancel out, leaving $A_{\rho_5} = \rho_5(s)$.

6-3. Eigenvalues

$\det(A_{\rho_5} - \lambda I) = (1-\lambda)(-1-\lambda) = 0 \Rightarrow \lambda = 1, -1$ (each mult. 1).
In the full matrix A, with $\deg(\rho_5) = 2$:

Eigenvalue	Mult. in A_{ρ_5}	Mult. in A
1	1	$1 \times 2 = 2$
-1	1	$1 \times 2 = 2$

VII. Full $\mathrm{Spec}(A)$ and $\mathrm{Spec}(L)$

Adjacency Spectrum

Eigenvalue λ	Source ρ	Multiplicity in A
3	ρ_1 (Trivial)	1
1	ρ_2	1
1	ρ_5	2
-1	ρ_3	1
-1	ρ_5	2
-3	ρ_4	1

Merged: $\mathrm{Spec}(A) = \{3^{(1)}, 1^{(3)}, -1^{(3)}, -3^{(1)}\}$

Laplacian Spectrum ($\mu = 3 - \lambda$)

$$\mathrm{Spec}(L) = \{0^{(1)}, 2^{(3)}, 4^{(3)}, 6^{(1)}\}$$

VIII. Comparison Table

Case	Surface	Min Non-trivial λ	Multiplicity	Source ρ	Min Non-trivial μ	Multiplicity
D_4, sphere	$\chi = 2$	1	3	$\rho_2 \oplus \rho_5$	2	3
D_4, torus	$\chi = 0$	1	3	$\rho_2 \oplus \rho_5$	2	3
Q_8, torus	$\chi = 0$	0	6	$\rho_2 \oplus \rho_3 \oplus \rho_5$	4	6

IX. Statement of Conjecture

> **If the face system of a Cayley graph forms a torus $\Rightarrow$ Multiplicity of min non-trivial eigenvalue of $\mathrm{Spec}(A$**

Intuition: A torus has two independent winding directions (horizontal and vertical), corresponding to two independent translation generators. These directions each contribute a dimension to the eigenspace, making the multiplicity at least 2.

Note: Since the D_4 sphere and torus use the same Cayley graph, their $\mathrm{Spec}(A)$ are identical. The conjecture's distinction lies in which eigenspaces are "utilized" by the face system. This detail must be clarified in the next phase.